

TEMPORAL IDENTIFYING OF INFLUENTIAL NODES IN COMPLEX CITATION NETWORKS

25-1-R-18

Project goals:

- Using gravity models to analyze papers behavior in citation networks.
- Develop a tool to analyze how the influence of each paper changes over the years.
- Research must contain different Gravity models and datasets.

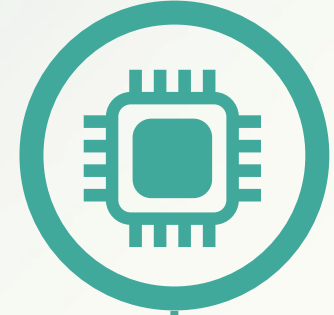
Selected models:

- Local Gravity Model - limits the depth of neighbor consideration using a truncation radius.
- Weighted Gravity Model - scales gravity calculations using values from the eigenvector corresponding to the largest eigenvalue of the adjacency matrix.
- Effective Distance Gravity Model - uses effective distance to calculate the distance between nodes.
- PageRank Gravity Model - scales gravity calculations using PageRank scores.



Choosing citation network

Searching the internet for datasets that are both large enough to produce results, while also being computable by chosen models of different complexity on standard hardware.



Data preprocessing

The system is designed to be compatible with models that require graph transpose. In addition, the creation of a topologically ordered graph by publication year is required. As a result, nodes have to be sorted.



Influence score calculation

Implementing models with optimization to reduce code complexity: using sparse matrices, libraries with built-in optimization, and reducing computation.



Result analysis

The analysis is based on derivative calculations to identify significant papers, such as those with the highest or most consistent growth.



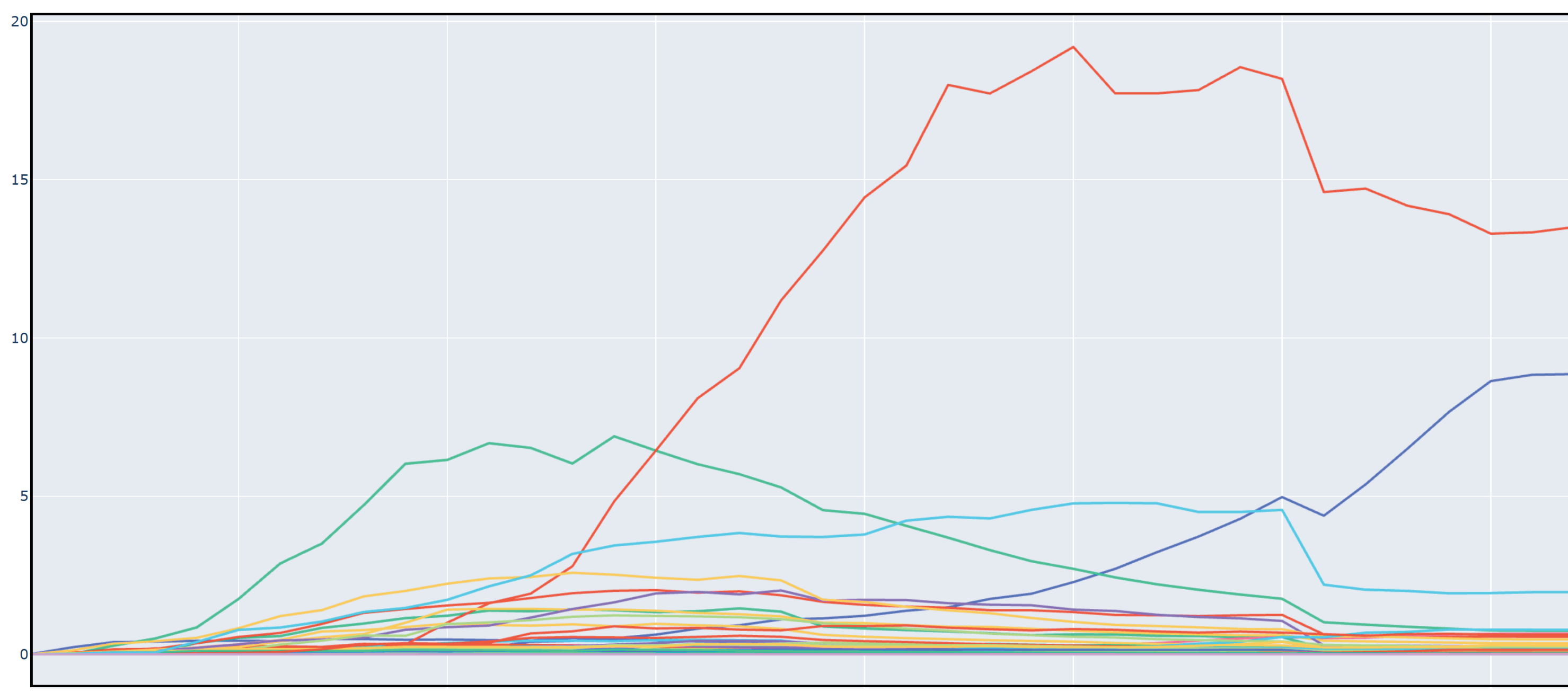
Display the results

The system is designed to work with thousands of tracked papers by filtering them using different criteria before plotting.

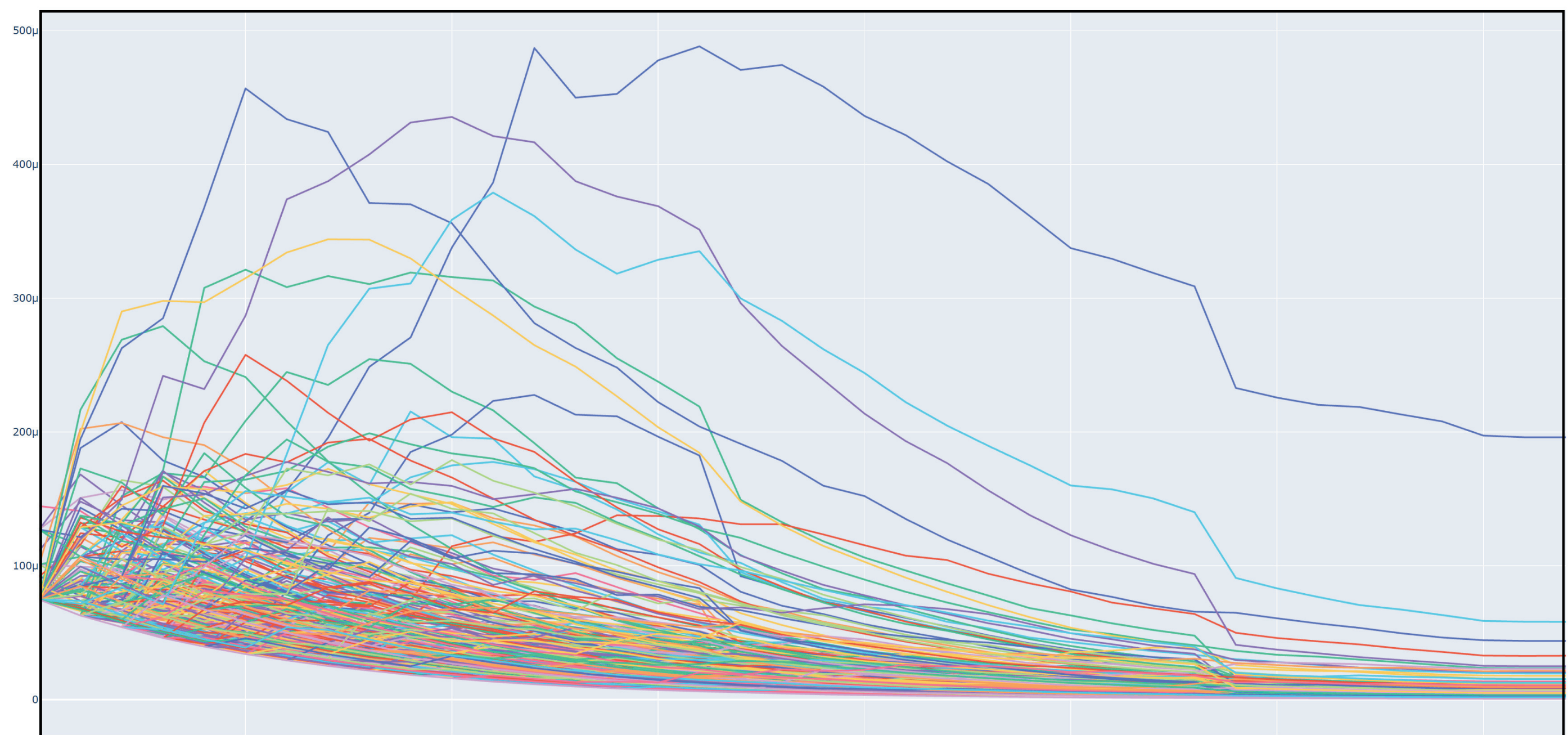


Manual analysis

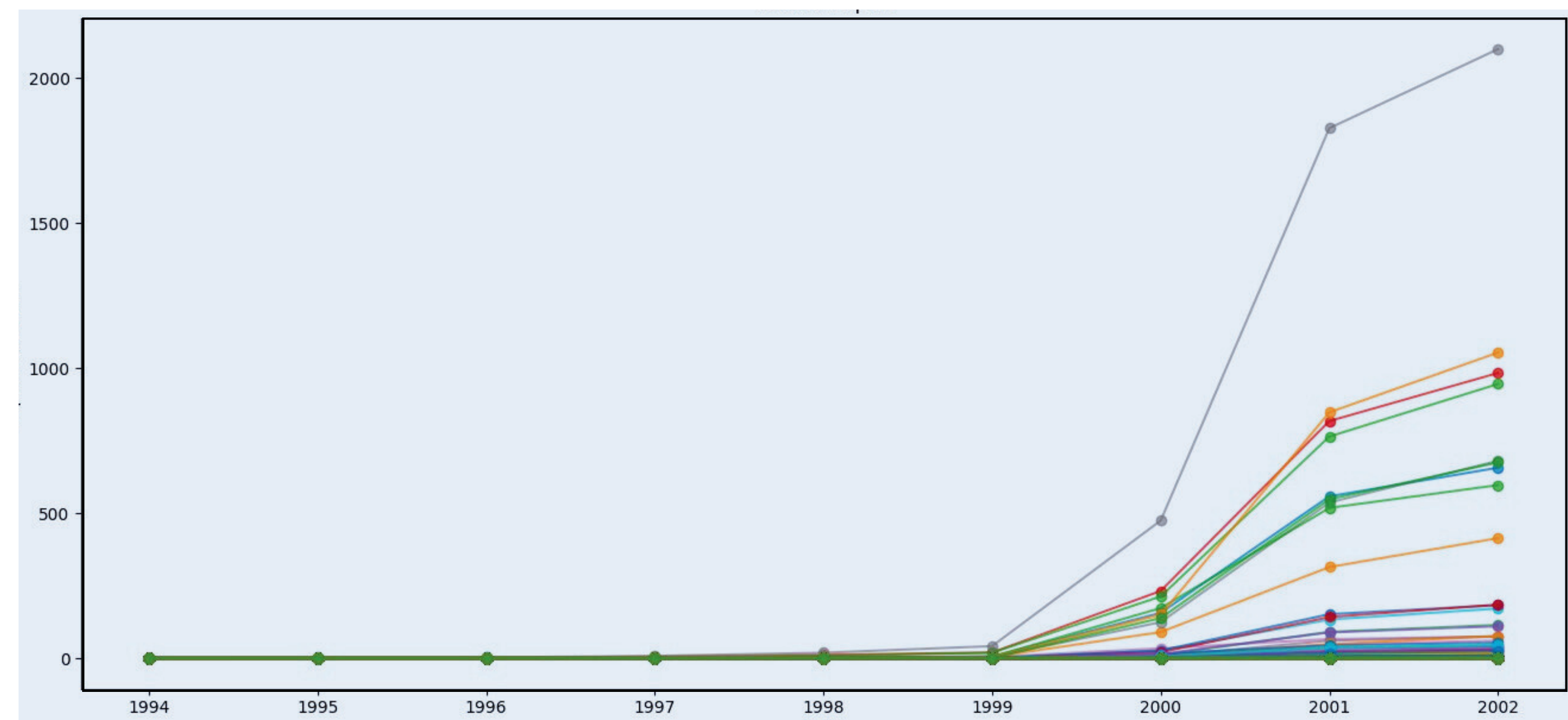
Utilize Plotly in the browser to dynamically hide papers. Apply standard deviation and result normalization for analysis.



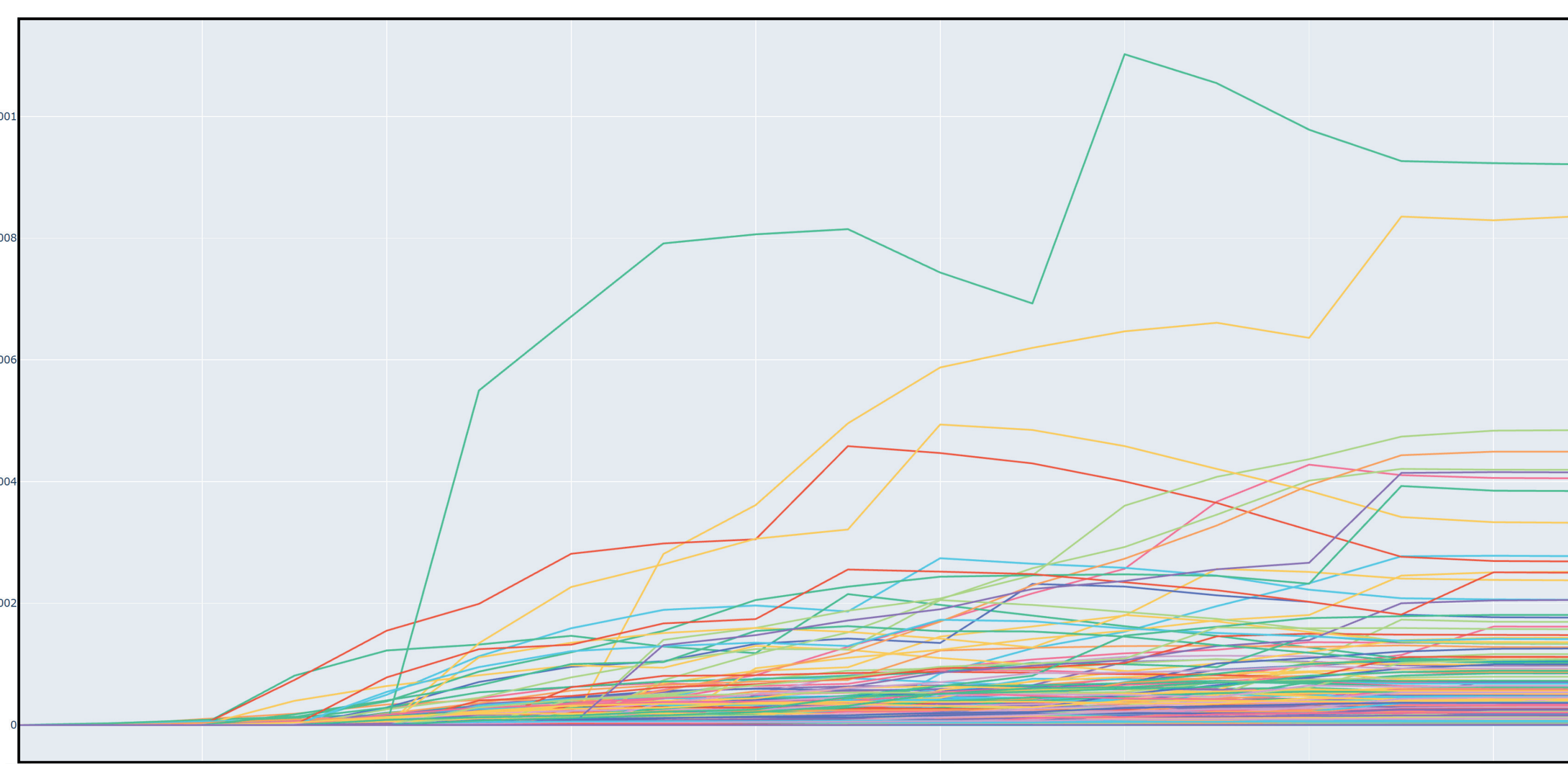
PageRank Gravity model, 1980 year of interest



PageRank model, 1980 year of interest



Effective Distance Gravity model, 1994 year of interest



Local Gravity model, 2000 year of interest

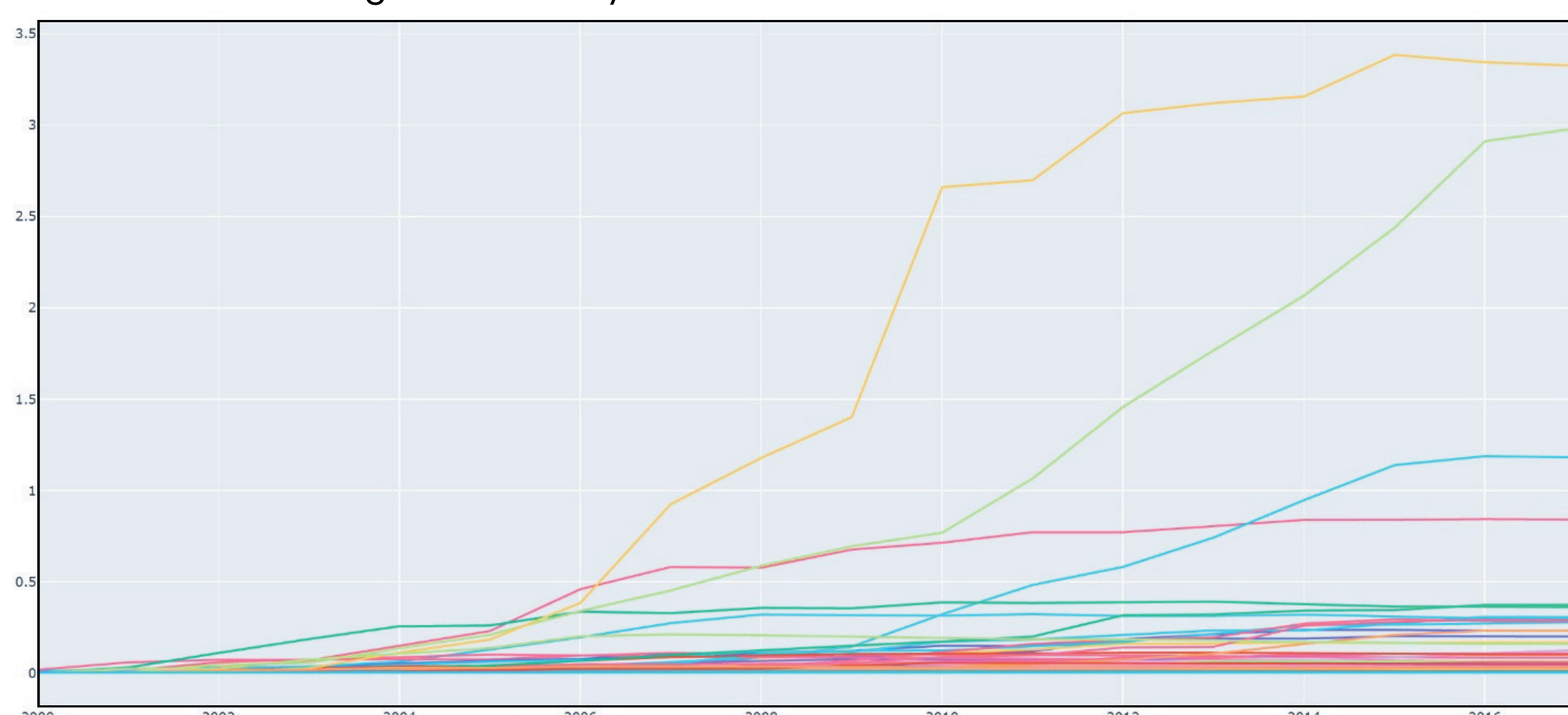
Practical implementations:

- A tool for evaluating a scientist's eligibility for awards by measuring the impact of their publications on the field.
- Identifying scientific trends, breakthrough ideas, or shifts in research focus to guide future studies.
- Helping the startups to forecast the direction of science and allocate resources effectively (e.g., funding, collaborations, and investments in specific fields).
- Improving academic learning program in higher education by taking in account the most relevant and impactful scientific publications.

Our research reveals a consistent pattern in the behavior of scientific papers:

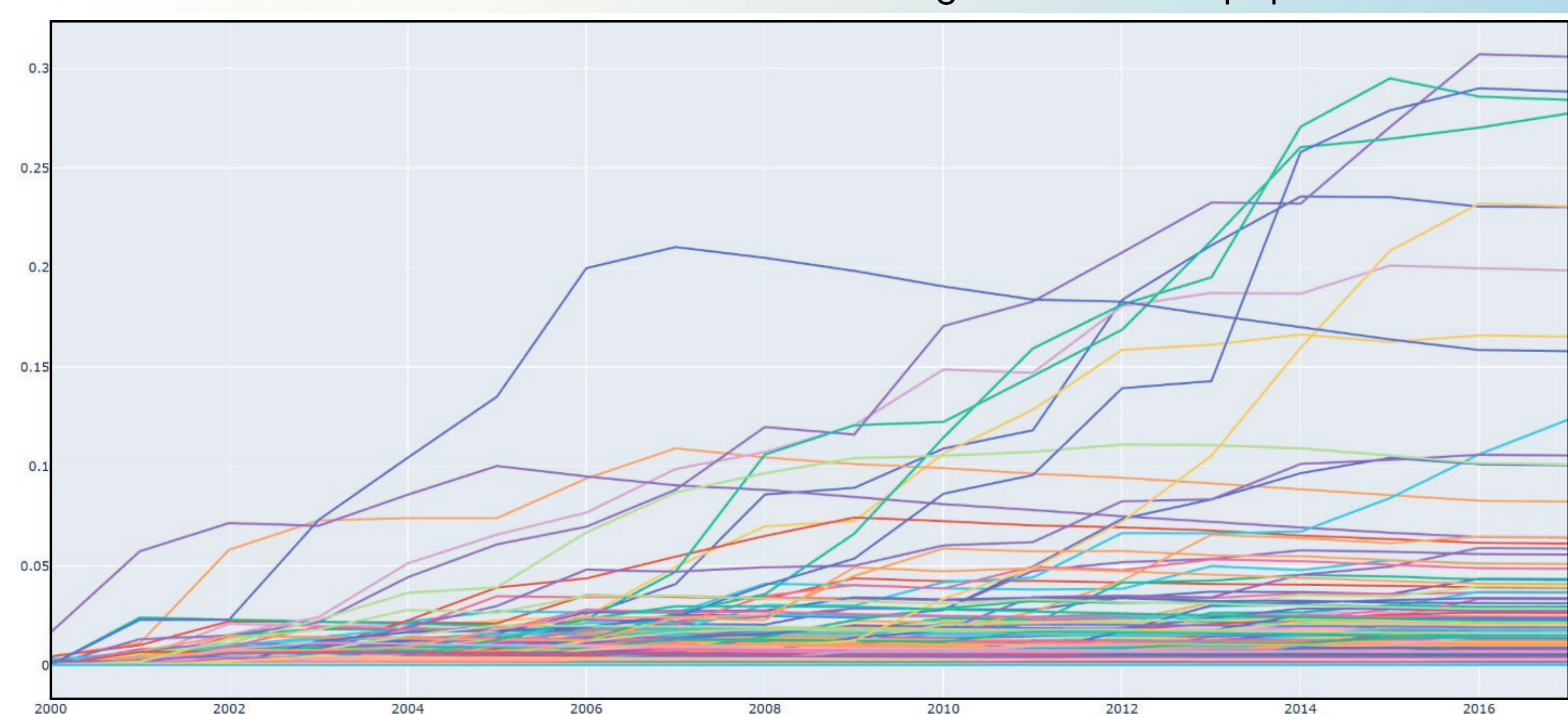
The majority of papers tend to form a dense cluster, and this trend appears to be constant, while only a few papers stand out and can be considered influential or interesting. For example, in a sample of around 10,000 papers, less than 1% receive high scores. Since these results are consistently observed across different datasets and various gravity models, this behavior is likely a characteristic of scientific publications.

Execution of the PageRank Gravity model on DBLP dataset



PageRank Gravity model 2000 year of interest

Another execution of the same model after removing the successful papers



PageRank Gravity model, 2000 year of interest, after excluding successful papers